Los Angeles Crime Data Analysis: 2020–Present

A data-driven investigation of 1M+ incidents — temporal patterns, geographic concentration, crime types, and actionable intelligence for technical stakeholders.

1.005M Incidents Analyzed

Dataset Scale

1,005,198 individual crime records spanning report dates from 2020 through early 2025

Analytical Dimensions

24 original fields reduced to 15 analytical dimensions after cleaning and feature engineering

Data Integrity

Zero duplicates confirmed. DR_NO primary key integrity fully validated across all records

Data Pipeline: Python → PostgreSQL → Power BI

Ingest & Clean

Python: Pandas, NumPy; null profiling

Store

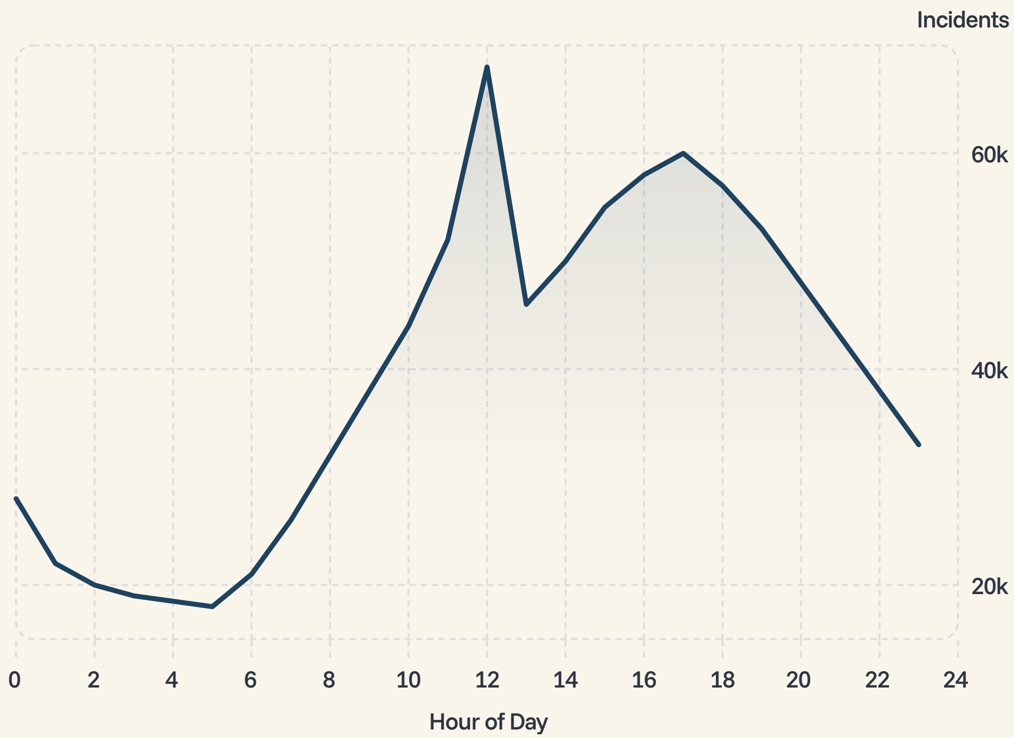
PostgreSQL: raw & cleaned tables, audit trail

Visualize

Power BI: 5 drill-down analytics pages

Columns missing more than 50% of their values were dropped programmatically. Missing Vict Sex and Premis Desc values were filled with the statistical mode of each column. Hour and Age Group features were engineered to enable temporal and demographic segmentation.

The Noon Peak: 68K Crimes at Hour 12



Reading the Signal

Midday Surge

Peak at 12:00 reaches **68,000 incidents** — 3.8× the overnight trough of 18,000 at hour 5

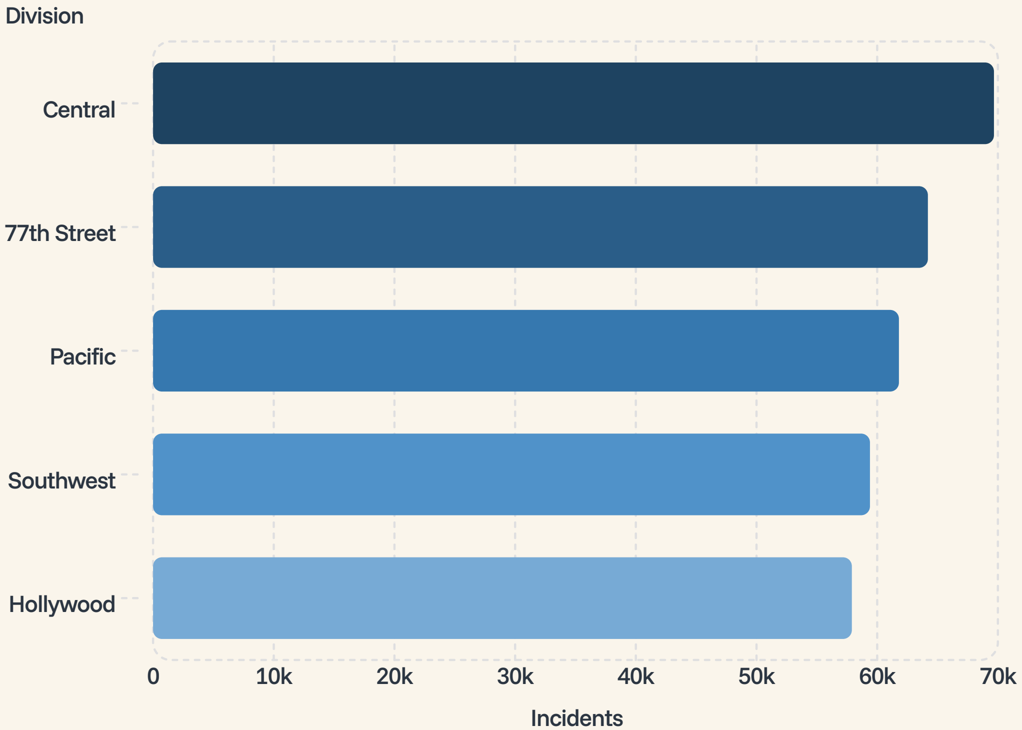
Mixed Signal

Noon spike reflects both **genuine behavioral concentration** and data-entry convention using 12:00 as a default for unknown occurrence times

Authentic Pattern

53K–60K incidents from 15:00–20:00 represents true behavioral signal and should anchor patrol scheduling decisions

Geographic Concentration: 5 Areas Dominate



What the Distribution Tells Us

Central Division leads with 69,674 incidents — 6.9% of the entire citywide total concentrated in a single patrol area.

The **top 5 divisions** — Central, 77th Street, Pacific, Southwest, and Hollywood — account for roughly **35% of all incidents**.

The remaining **16 divisions** share ~65% of incidents, a distribution with direct implications for patrol density, resource staffing models, and budget allocation across the department.

> Top 5 of 21 divisions generate over one-third of all citywide crime volume.

Vehicle Theft Dominates the Crime Mix

115K

Vehicle – Stolen

11.5% of all crimes — single largest offense category in the dataset

74.8K

Battery / Simple Assault

Second most frequent offense type across all five years

63.5K

Burglary From Vehicle

Closely tied to vehicle theft — parking infrastructure as a crime nexus

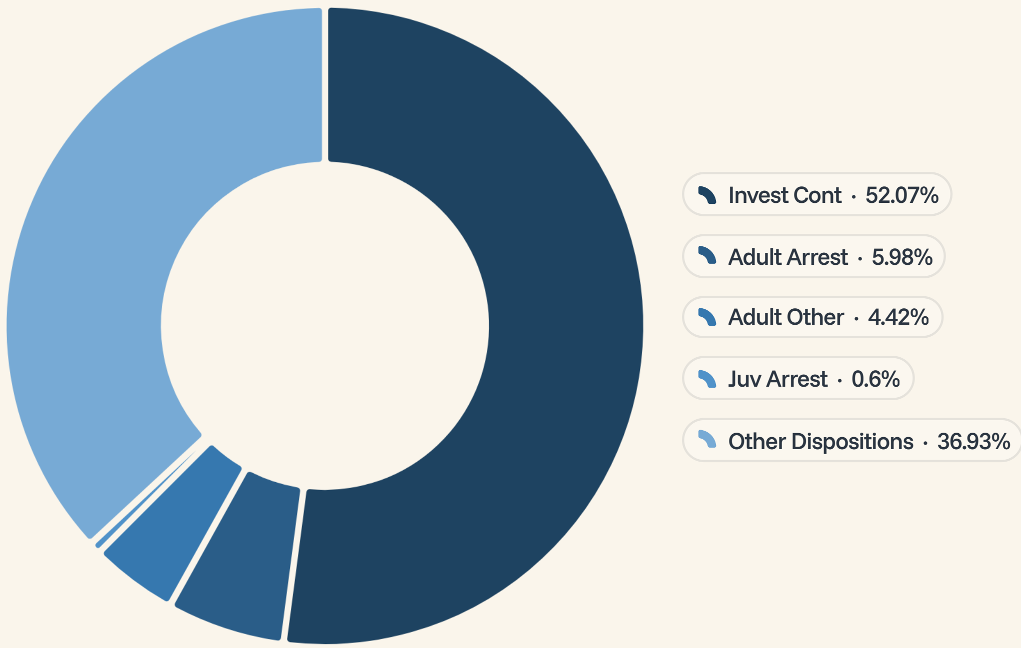
37.5%

Top 5 Share

Vehicle theft, battery, burglary, identity theft, and vandalism account for 377K of 1M incidents

With 140 distinct crime types recorded, the distribution features a steep power-law curve — the top 5 categories account for over a third of all incidents, while the long tail contains hundreds of rare offense subtypes.

Case Status: 83% Under Investigation



Disposition Breakdown

Invest Cont — 523,420

Cases remain open under continued investigation — the dominant disposition by a wide margin

Adult Arrest — 60,160

A low closure rate relative to the sheer volume of reported incidents

Adult Other — 44,420

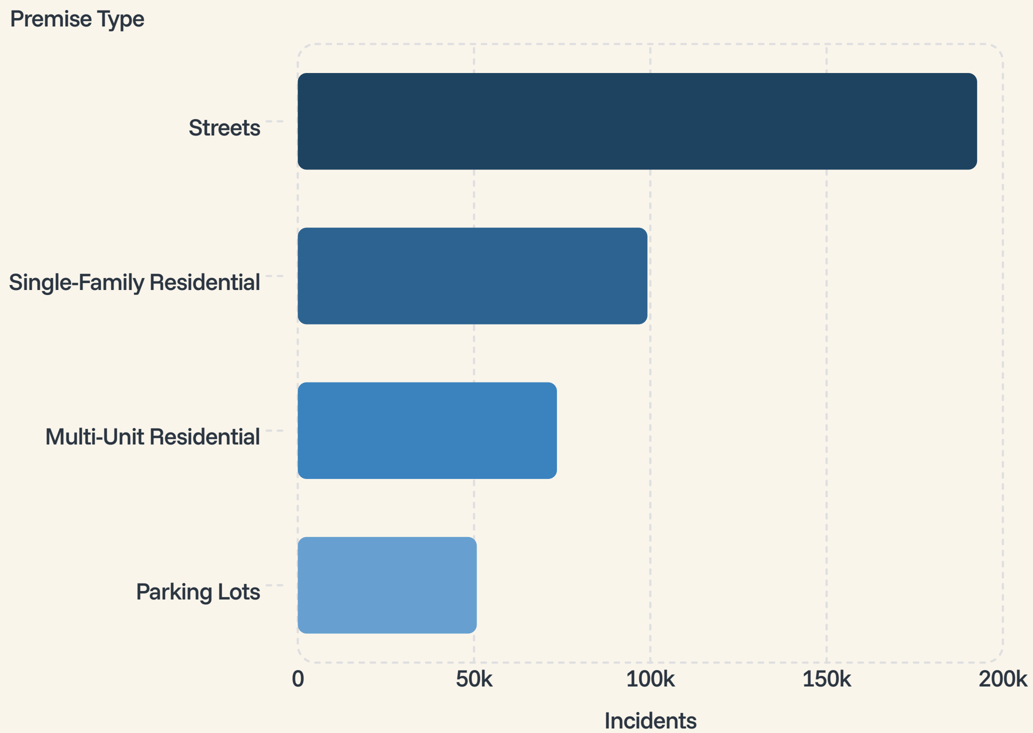
Cases resolved through alternative adult dispositions outside formal arrest

Juv Arrest — 6,020

Minimal juvenile involvement in the overall reported crime pool

Premises & Temporal Patterns

Where Crimes Occur



When Crimes Occur

Streets account for nearly 1-in-5 incidents. Friday–Saturday weekend elevation and the January winter surge are robust patterns suitable for staffing optimization models.

Time of Day

Night 278.61K > Afternoon 268.58K > Morning 227.24K

Day of Week

Friday highest (153.72K) vs. Sunday lowest (139.66K) — **10% variance**

Seasonal Pattern

January peak (92.78K/month) vs. November–December dip (78K) — **18% variance**

Actionable Insights & Next Steps

01

Resource Allocation

Concentrate patrol resources in Central, 77th Street, Pacific, Southwest, and Hollywood divisions — especially during **Friday evenings and the 15:00–20:00 window**. January staffing should account for the seasonal crime surge.

02

Crime Prevention Targeting

Vehicle theft prevention programs (115K incidents) offer the **highest single-category impact**. Parking infrastructure — lots accounting for 50.8K incidents — is a concentrated intervention point. Adults aged 31–40 represent the primary victim demographic for outreach programs.

03

Data Governance

Standardize occurrence timestamp entry to reduce noon-spike artifact contamination. Enforce victim age recording for property crimes to eliminate zero-value imputation bias. Both improvements are **prerequisites for reliable predictive modeling** on this dataset.

Conclusion: Data-Driven Public Safety in Action

The Evidence is Clear

LA's crime landscape is structured by geography, time, and offense type. Five divisions account for **35% of all incidents**; vehicle theft represents **11.5%** of crimes; the 31–40 demographic comprises **45.5%** of all victims.

The Opportunity is Actionable

Temporal patterns — Friday peaks, 15:00–20:00 window, January surge — and geographic concentration enable **precision resource deployment** and highest single-category impact through vehicle theft prevention.

The Path is Clear

Implement data quality standards. Deploy predictive models for patrol scheduling. Establish divisional dashboards for real-time monitoring. Target victim outreach to **high-risk demographics**.

The Impact is Measurable

Success metrics: incident reduction, response time improvement, **case closure rate gains**, and resource optimization across all 21 divisions.

> **The data is clear. The path is actionable. The time to act is now.** By leveraging this 1M-incident dataset with precision, LAPD can prevent crimes before they occur, optimize resource deployment, and serve the Los Angeles community more effectively than ever before.